

[TODO: Paper title (work package WP-NN)]

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Abstract

[TODO: Abstract. Problem in one sentence. Approach in one sentence. Main theorem or measurement, with its preconditions named in the same sentence. One sentence of evaluation evidence. One sentence of scope or limitation.]

1 Introduction

[TODO: Motivation grounded in the deployed-systems problem, not the framework. Arrive as an implementer with running artifacts and measurements; the model earns its place by what it predicts.]

[TODO: Statement of the main result, informally, with preconditions named.]

Contributions.

1. [TODO: Contribution 1 (Section 4).]
2. [TODO: Contribution 2 (Section 5).]
3. [TODO: Contribution 3.]

2 Related Work

Information-theoretic secrecy. [TODO: Position against the wiretap channel [Wyn75], its Gaussian analysis [LYCH78], and broadcast channels with confidential messages [CK78]; foundations in [Sha48, Fan61, CT06].]

Quantitative information flow. [TODO: Covert-channel capacity [Mil87]; language-based information-flow security [SM03]; differential privacy as the neighbouring guarantee family [DR14].]

Multi-agent LLM leakage. [TODO: Inter-agent channel measurement [EYBMAM26]; compositional leakage bounds for sequential agent pipelines [AA26].]

Estimation and verification tooling. [TODO: Mutual-information estimation [KSG04, BBR⁺18]; succinct arguments where attestation is relevant [Gro16, GWC19, KST22].]

3 Model and Preliminaries

3.1 Notation

[TODO: Random variables, alphabets, entropy $H(X)$, mutual information $I(X;Y)$, conditioning conventions. Cite [CT06] for standard definitions rather than restating them.]

3.2 System model

[TODO: Agents, channels, and composition. Name each architectural precondition as a numbered Assumption so later theorems can reference it.]

Assumption 3.1 ([TODO: name]). [TODO: State the precondition precisely, including the adversary class it holds against.]

3.3 Threat model

[TODO: Adversary capabilities, what is observable, what is excluded, and why the exclusions are realistic for the deployment class studied.]

4 Main Results

Theorem 4.1 ([TODO: name]). *Under Assumption 3.1, [TODO: statement].*

Proof. [TODO: Proof, or a sketch with the full proof in the appendix.] □

5 Evaluation

5.1 Setup

[TODO: Benchmark or measurement design; scenario counts, models, trace counts. Where the design extends a published benchmark, cite it and state exactly what changed [EYBMAM26].]

5.2 Estimators

[TODO: Mutual-information estimation method and its bias characteristics [KSG04, BBR⁺18].]

5.3 Results

[TODO: Tables and figures. Report effect sizes with uncertainty; compare measured leakage against the bound of Theorem 4.1 and against published baselines [AA26].]

Artifact availability. [TODO: Where the harness, traces, and analysis code are published.]

6 Threats to Validity

Construct validity. [TODO: Does the measured quantity operationalise the claimed one? Estimator bias, proxy gaps.]

Internal validity. [TODO: Confounds in the experimental design; sensitivity analyses run.]

External validity. [TODO: Which deployment classes the results transfer to, and which they do not. Adversary classes excluded by assumption.]

Assumption failure. [TODO: For each numbered Assumption: the consequence of its violation, stated quantitatively where possible (e.g. collusion between agents collapses the conditional-independence residual that the bound relies on).]

7 Conclusion

[TODO: Two or three paragraphs: result, scope, open problems.]

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